



Elementary Science

Light Sources

Level 1

Topic 9: Light & Sound | Worksheet 1

Name: _____

Date: _____

A Match each object to its group.

Draw a line to LIGHT SOURCE or NOT A LIGHT SOURCE.



Sun

•

NOT a source



Torch

•



Candle

•



Moon

•

LIGHT SOURCE



Mirror

•



Light bulb

•

B Circle the best answer.

1. Which gives off its own light?

sun

moon

mirror

2. Which reflects light but does not make it?

candle

moon

bulb

3. Which can help us see in a dark room?

torch

book

shoe

4. Which is a natural light source?

lamp

Sun

TV

C Fill in the blanks.

Word bank: source reflect dark light

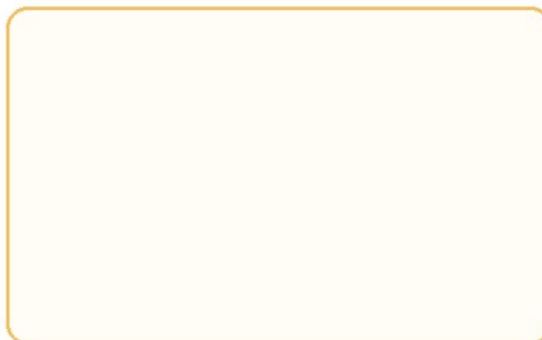
1. The Sun is a natural light _____.

2. We need _____ to see objects.

3. The Moon can _____ sunlight.

4. A torch helps us see in the _____.

D Challenge: Draw another light source



Why is it a light source?

Light and sound help us explore the world!



Elementary Science

Shadows

Level 2

Topic 9: Light & Sound | Worksheet 2

Name: _____

Date: _____

A Match each word to its meaning.

shadow

lets most light pass through

opaque

dark shape made when light is blocked

transparent

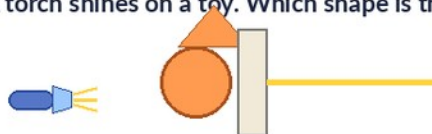
gives off light

light source

blocks light

B Choose the correct shadow.

A torch shines on a toy. Which shape is the shadow?



A



B



C



The shadow has the same outline as the object blocking the light.

C Complete the sentences.

Word bank: blocked longer shorter light

1. A shadow forms when _____ is blocked.
2. An opaque object _____ light.
3. A low Sun can make shadows look _____.
4. When the Sun is high, shadows are often _____.

D Explain: Why do shadows change size

Think about where the light source and object are.

Light and sound help us explore the world!



Elementary Science

How Sound Is Made

Level 3

Topic 9: Light & Sound | Worksheet 3

Name: _____

Date: _____

A Match the word to its meaning.

vibration



how high or low a sound is

pitch



back-and-forth movement

volume



how loud or quiet a sound is

sound wave



carries sound energy

B What is vibrating?

1. Drum



vibrates: _____

2. Guitar



vibrates: _____

3. Speaker



vibrates: _____

4. Tuning fork



vibrates: _____

C True or False?

1. Sound is made by vibrations.

True

False

2. A louder sound always has a high

True

False

3. Sound can travel through air.

True

False

4. If something stops vibrating, its

True

False

D Explain: How can we change a sound?

Give one way to change:

1. volume (loud/quiet)

2. pitch (high/low)

Sound begins when something vibrates and sends sound waves through a material such as air.

Light and sound help us explore the world!



Elementary Science

Light & Sound Investigation

Level 4

Topic 9: Light & Sound | Worksheet 4

Name: _____

Date: _____

A Shadow test: Look at the results.

Same torch and object; only the distance changes.

Object position	Shadow size	Result
near torch	_____	A
middle	_____	B
near screen	_____	C

What happened as the object moved toward the screen?

B Sound test: Look at the results.



Same speaker volume,
listener at different distances.

Distance	Sound	Result
1 m	_____	A
3 m	_____	B
5 m	_____	C

What happened as the listener moved farther away?

C Answer the questions.

1. Which position made the largest shadow?

2. Which distance sounded loudest?

3. What should stay the same in a fair test?

4. Why repeat a test?

D Challenge: What did we learn?

Write one conclusion about light and one about sound.

Light:

Sound:

Investigate, observe, and explain like a scientist!



Elementary Science

Light Sources

Level 1

Topic 9: Light & Sound | Answer Sheet 1

Name: _____

Date: _____

A Match each object to its group.

Draw a line to LIGHT SOURCE or NOT A LIGHT SOURCE.



Sun



Torch



Candle



Moon



Mirror



Light bulb

NOT a source

LIGHT SOURCE

B Circle the best answer.

1. Which gives off its own light?

sun

moon

mirror

2. Which reflects light but does not make it?

candle

moon

bulb

3. Which can help us see in a dark room?

torch

book

shoe

4. Which is a natural light source?

lamp

Sun

TV

C Fill in the blanks.

Word bank: source reflect dark light

1. The Sun is a natural light _____. **source**

2. We need _____ to ~~light~~ objects.

3. The Moon can _____ sunlight. **reflect**

4. A torch helps us see in the _____ **dark**

D Challenge: Draw another light source



Example: the Sun

Why is it a light source?

It gives off its own light.

Light and sound help us explore the world!

Shadows — Answer Key

Light and Sound • Level 2

A. Match each word to its meaning.

1. shadow

Answer: a dark shape made when light is blocked

2. opaque

Answer: blocks light

3. transparent

Answer: lets most light pass through

4. light source

Answer: gives off light

B. Choose the correct shadow for the toy shown (a circle with a triangle on top).

1. Which shape is the toy's shadow — A (plain circle), B (keyhole shape), or C (circle with a triangle on top)?

Answer: C — a shadow's outline always matches the outline of the object blocking the light.

C. Complete the sentences (word bank: blocked, longer, shorter, light).

1. A shadow forms when _____ is blocked.

Answer: light

2. An opaque object _____ light.

Answer: blocks (blocked)

3. A low Sun can make shadows look _____.

Answer: longer

4. When the Sun is high, shadows are often _____.

Answer: shorter

D. Why do shadows change size?

Answer: Shadow size depends on where the light source is. When the light is low or close to an object, the shadow stretches out and looks longer; when the light is high overhead, the shadow is shorter. Moving an object closer to the light also makes its shadow bigger.



Elementary Science

How Sound Is Made

Level 3

Topic 9: Light & Sound | Answer Sheet 3

Name: _____

Date: _____

A Match the word to its meaning.

vibration

how high or low a sound is

pitch

back-and-forth movement

volume

how loud or quiet a sound is

sound wave

carries sound energy

B What is vibrating?

1. Drum



vibrates: skin

2. Guitar



vibrates: string

3. Speaker



vibrates: cone

4. Tuning fork



vibrates: metal arms

C True or False?

1. Sound is made by vibrations.

☒ True

☐ False

2. A louder sound always has a higher pitch.

☐ True

☒ False

3. Sound can travel through air.

☒ True

☐ False

4. If something stops vibrating, its sound stops.

☒ True

☐ False

D Explain: How can we change a sound?

Give one way to change:

1. volume (loud/quiet)

hit/pluck more strongly

2. pitch (high/low)

change how fast it vibrates

Sound begins when something vibrates and sends sound waves through a material such as air.

Light and sound help us explore the world!



Elementary Science

Light & Sound Investigation

Level 4

Topic 9: Light & Sound | Answer Sheet 4

Name: _____

Date: _____

A Shadow test: Look at the results.

Same torch and object; only the distance changes.

Object position	Shadow size	Result
near torch	large	A
middle	medium	B
near screen	small	C

What happened as the object moved toward the screen?

The shadow became smaller.

B Sound test: Look at the results.



Same speaker volume,
listener at different distances.

Distance	Sound	Result
1 m	loud	A
3 m	medium	B
5 m	quiet	C

What happened as the listener moved farther away?

The sound became quieter.

C Answer the questions.

1. Which position made the largest shadow?

near the torch

2. Which distance sounded loudest?

1 m

3. What should stay the same in a fair test?

all other conditions

4. Why repeat a test?

to check the result is reliable

D Challenge: What did we learn?

Write one conclusion about light and one about sound.

Light:

Moving an object changes the size of its shadow.

Sound:

Sound becomes quieter as you move farther from the source.

Investigate, observe, and explain like a scientist!